Assignment 2: Identify Common Sensors Used in IoT Projects

Internship: IoT with AI

Week: 1

Topic: Introduction to IoT and Electronics Basics

1. Introduction

Sensors are important components of Internet of Things (IoT) systems. They detect and measure physical conditions from the surrounding environment and convert them into electrical signals or digital data.

These readings are collected by microcontrollers such as Arduino and ESP32. The collected data can be processed and transmitted through the internet for monitoring, analysis and automation.

2. Common Sensors Used in IoT Projects

1. Temperature Sensor

A temperature sensor measures the temperature of the surrounding environment or an object.

Applications:

- Weather monitoring.
- Smart home temperature control.
- Industrial temperature monitoring.
- Greenhouse automation.

Example: LM35 and TMP36.

2. Humidity Sensor

A humidity sensor measures the amount of moisture present in the air.

Applications:

- Weather stations.
- Smart agriculture.
- Greenhouses.
- Indoor air monitoring.

Example: DHT11 and DHT22.

3. Ultrasonic Sensor

An ultrasonic sensor measures distance by transmitting ultrasonic sound waves and calculating the time taken for the reflected waves to return.

Applications:

- Obstacle detection.
- Smart parking systems.
- Distance measurement.
- Automatic water-level monitoring.

Example: HC-SR04.

4. PIR Motion Sensor

A Passive Infrared (PIR) sensor detects movement by sensing changes in infrared radiation emitted by objects.

Applications:

- Motion-based security systems.
- Automatic lighting.
- Intruder detection.
- Smart home automation.

Example: HC-SR501 PIR sensor.

5. Light Sensor

A light sensor detects the intensity of light in its surroundings.

Applications:

- Automatic streetlights.
- Smart lighting systems.
- Solar tracking systems.
- Automatic brightness adjustment.

Example: LDR (Light Dependent Resistor).

6. Gas Sensor

A gas sensor detects the presence or concentration of particular gases in the surrounding environment.

Applications:

- Gas leakage detection.
- Industrial safety systems.
- Air quality monitoring.

- Fire and hazard warning systems.

Example: MQ-2 gas sensor.

7. Soil Moisture Sensor

A soil moisture sensor measures the moisture content present in the soil.

Applications:

- Smart irrigation.
- Agricultural monitoring.
- Automatic watering systems.
- Plant monitoring.

Example: Capacitive soil moisture sensor.

8. Sound Sensor

A sound sensor detects sound waves and converts them into electrical signals.

Applications:

- Noise monitoring.
- Voice-activated systems.
- Security alarms.
- Sound-based automation.

Example: KY-038 sound sensor module.

9. Pressure Sensor

A pressure sensor measures the pressure of gases or liquids.

Applications:

- Weather monitoring.
- Industrial process control.
- Water pressure monitoring.
- Automotive systems.

Example: BMP280 pressure sensor.

10. Accelerometer Sensor

An accelerometer measures acceleration and detects changes in movement, tilt and orientation.

Applications:

- Fitness tracking.
- Smart wearables.
- Vehicle motion detection.
- Vibration monitoring.

Example: MPU6050 accelerometer module.

3. Summary Table

Sensor| Measures or Detects| Example

Temperature| Temperature| TMP36

Humidity| Air moisture| DHT11

Ultrasonic| Distance| HC-SR04

PIR| Motion| HC-SR501

Light| Light intensity| LDR

Gas| Gas concentration| MQ-2

Soil Moisture| Soil water content| Capacitive sensor

Sound| Sound intensity| KY-038

Pressure| Pressure| BMP280

Accelerometer| Acceleration| MPU6050

4. Importance of Sensors in IoT

1. Sensors collect real-time environmental data.

2. They help automate different processes.

3. They improve monitoring and control.

4. They reduce manual intervention.

5. They support data-driven decision-making.

6. They help develop smart and connected systems.

5. Conclusion

Sensors are essential components of IoT systems because they collect information from the physical environment. Different sensors are used to measure temperature, humidity, distance, light, motion, pressure and other physical parameters. When connected to microcontrollers such as Arduino and ESP32, these sensors help develop intelligent monitoring and automation systems.